Field of Data: Growing Insights from Smart Farming

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Data Overview

Smart Data Farming 2024 (SF24)

- Consists of farming data from California farms in 2024
- 2200 entires, 28 features (23 original, 5 derived)

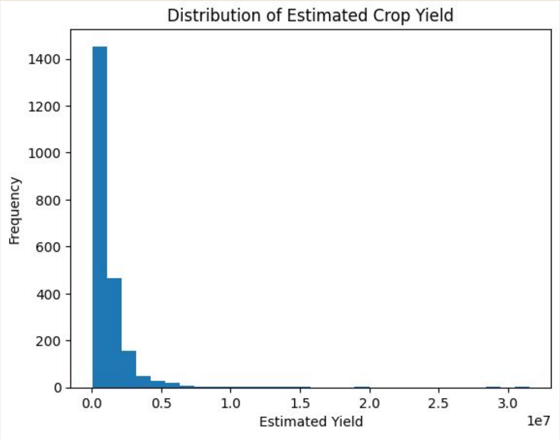
https://www.kaggle.com/datasets/datasetengineer/smart-farming-data-2024-sf24/data

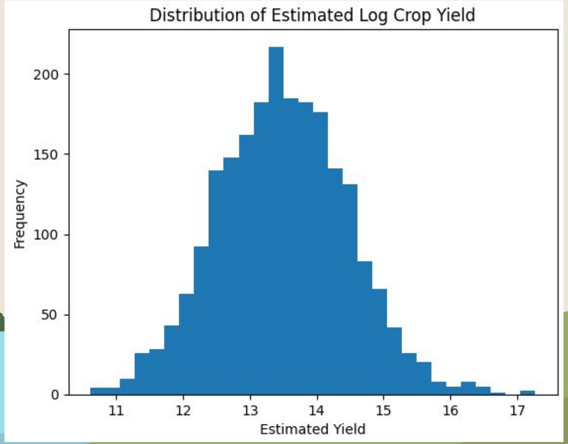
	N	P	K	temperature	humidity	ph	rainfall	label	soil_moisture	soil_type	...	organic_matter	irrigation_frequency	crop_density	pest
0	90	42	43	20.879744	82.002744	6.502985	202.935536	rice	29.446064	2	...	3.121395	4	11.743910	
1	85	58	41	21.770462	80.319644	7.038096	226.655537	rice	12.851183	3	...	2.142021	4	16.797101	
2	60	55	44	23.004459	82.320763	7.840207	263.964248	rice	29.363913	2	...	1.474974	1	12.654395	
3	74	35	40	26.491096	80.158363	6.980401	242.864034	rice	26.207732	3	...	8.393907	1	10.864360	
4	78	42	42	20.130175	81.604873	7.628473	262.717340	rice	28.236236	2	...	5.202285	3	13.852910	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
2195	107	34	32	26.774637	66.413269	6.780064	177.774507	coffee	10.697757	1	...	4.720355	5	18.597260	
2196	99	15	27	27.417112	56.636362	6.086922	127.924610	coffee	12.203830	3	...	4.141148	6	15.417979	
2197	118	33	30	24.131797	67.225123	6.362608	173.322839	coffee	28.989176	3	...	1.599614	5	12.956675	
2198	117	32	34	26.272418	52.127394	6.758793	127.175293	coffee	13.642305	2	...	8.934077	6	16.868131	
2199	104	18	30	23.603016	60.396475	6.779833	140.937041	coffee	23.911728	3	...	6.401994	3	14.652123	

2200 rows × 23 columns

Cleaning/Derived Features

Feature Name	Description
estimatedAverageYieldPerPlant	Approximated yield per plant
cropYield	Estimated total crop yield **(crop density x avg yield per plant)**
logCropYield	Natural log of cropYield to reduce skewness
WAI (Water Availability Index)	Combines soil moisture and rainfall to assess overall water availability for crops **(soil_moisture + rainfall)**
PP (Photosynthesis Potential)	Estimates the potential for photosynthesis based on sunlight exposure, CO2 concentration, and temperature **(sunlight_exposure x CO2 x temperature)**
THI (Temperature-Humidity Index)	Assess the level of stress caused by heat and humidity combined **((0.5 x temperature) + 61.0 + ((temperature -68.0) * 1.2) + (humidity * 0.094))**
SFI (Soil Fertility Index)	Assesses soil fertility based on organic matter content and NPK levels **((N+P+K) x Organic Matter)**
NBR (Nutrient Balance Ratio)	Reflects the balance between nitrogen, phosphorus, and potassium in the soil **(Nitrogen + Phosphorus + Potassium / 3)**





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LASSO Regression

How do climate variables influence crop yield?

Lambda: 0.0018103420684136829
R^2: 0.438181989737757
Climate Variables:

- Temperature
- Humidity
- Rainfall
- Sunlight exposure
- Wind speed
- CO2 concentration
- Frost risk
- Temperature-humidity index
- Water availability index
- Photosynthesis potential

	feature	coefficient
0	temperature	0.204908
1	humidity	-0.055721
2	rainfall	-0.000000
3	sunlight_exposure	0.267975
4	wind_speed	0.005035
5	co2_concentration	0.057755
6	frost_risk	-0.352857
7	THI	0.000000
8	WAI	0.418683
9	PP	-0.000000

Which crops are most resilient under climate conditions?

Crops:

- apple
- banana
- blackgram
- chickpea
- coconut
- coffee
- cotton
- grapes
- jute
- kidneybeans
- lentil
- maize
- mango
- mothbeans
- mungbean
- muskmelon
- orange
- papaya
- pigeonpeas
- pomegranate
- rice
- watermelon

```
# of 0's for apple: 2
# of 0's for banana: 1
# of 0's for blackgram: 1
# of 0's for chickpea: 1
# of 0's for coconut: 0
# of 0's for coffee: 1
# of 0's for cotton: 3
# of 0's for grapes: 0
# of 0's for jute: 2
# of 0's for kidneybeans: 1
# of 0's for lentil: 2
# of 0's for maize: 1
# of 0's for mango: 3
# of 0's for mothbeans: 2
# of 0's for mungbean: 1
# of 0's for muskmelon: 1
# of 0's for orange: 0
# of 0's for papaya: 1
# of 0's for pigeonpeas: 2
# of 0's for pomegranate: 1
# of 0's for rice: 2
# of 0's for watermelon: 2
```

cotton: temperature, sunlight, CO_2

mango: CO_2, THI, PP

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Nutrient-Based Crop Classification

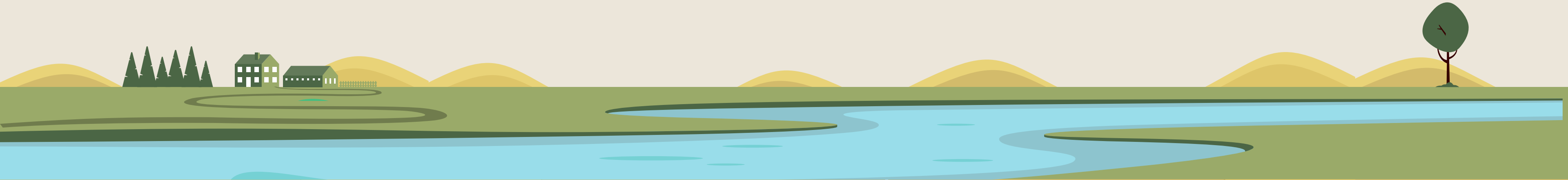
How can Soil Nutrient Patterns be used to identify Different Crops?

Why?

Different crops require different balances of Nitrogen, Phosphorus, and Potassium, analyzing nutrient patterns shows whether soil chemistry can meaningfully distinguish between crops.

Goal

Be able to distinguish crops solely on nutrient data.



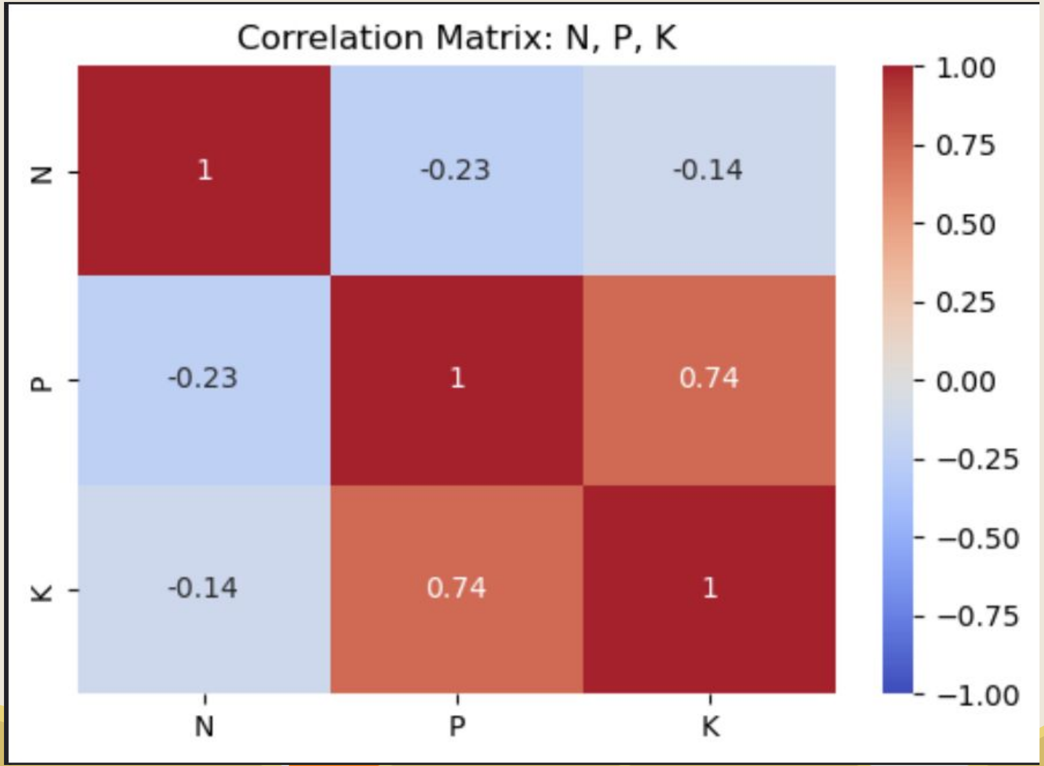
Key Data Used

Data Used:

N - Nitrogen content in soil (ppm)
K - Potassium content in soil (ppm)
P - Phosphorus content in soil (ppm)
Label - Crop Type (22 Unique)

Crops in Dataset:

- Rice
- Maize
- Chickpea
- Kidney Beans
- Pigeon Peas
- Moth Beans
- Mung Beans
- Blackgram
- Lentil
- Pomegranate
- Banana
- Mango
- Grapes
- Watermelon
- Muskmelon
- Apple
- Orange
- Papaya
- Coconut
- Cotton
- Jute
- Coffee


Correlation Matrix: N, P, K
N
P
K
1
-0.23
-0.14
-0.23
1
0.74
-0.14
0.74
1
N
P
K
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Initial Approach: Decision Tree

Why Model: Used for classification

Features: Nitrogen (N), Potassium (K), Phosphorous (P)

Target : Label (22 Crops)

Train/Test Split: 70/30

Max Depth: 4

Results:

- Model performed poorly and underfitted.
- Model was able to classify chickpea and bananas at a high level.
- Struggled with everything else

Accuracy: 0.2712121212121212124

Classification report:

	precision	recall	f1-score	support
apple	0.50	0.47	0.48	30
banana	0.94	1.00	0.97	30
blackgram	0.13	0.97	0.24	30
chickpea	1.00	1.00	1.00	30
coconut	0.00	0.00	0.00	30
coffee	0.00	0.00	0.00	30
cotton	0.48	1.00	0.65	30
grapes	0.50	0.53	0.52	30
jute	0.11	1.00	0.21	30
kidneybeans	0.00	0.00	0.00	30
lentil	0.00	0.00	0.00	30
maize	0.00	0.00	0.00	30
mango	0.00	0.00	0.00	30
mothbeans	0.00	0.00	0.00	30
mungbean	0.00	0.00	0.00	30
muskmelon	0.00	0.00	0.00	30
orange	0.00	0.00	0.00	30
papaya	0.00	0.00	0.00	30
pigeonpeas	0.00	0.00	0.00	30
pomegranate	0.00	0.00	0.00	30

How to Improve?

- Switch to Random Forest Model
- Grouped crops into families: Cereal, Legume, Fruits, and Industrial
- Include more features:
  - soil_type: type of soil (1= sandy, 2=Loamy, 3= Clay)
  - irrigation_frequency (times/week): Frequency of irrigation
  - Water_source_type: Type of water used for irrigation (1=river, 2 = Groundwater, 3= Recycled
  - ph: Soil ph level
- Created derived features for nutrient ratios:
  - N_to_P : Nitrogen to Phosphorus ratio
  - P_to_K: Phosphorus to Potassium ratio
  - N_to_K: Nitrogen to Potassium ratio

```
df['N_to_P'] = df['N'] / df['P']
df['P_to_K'] = df['P'] / df['K']
df['N_to_K'] = df['N'] / df['K']
```

```
cereals = ['rice', 'maize']
legumes = ['chickpea','kidneybeans','pigeonpeas','mothbeans','mungbean','blackgram','lentil']
fruits = ['banana','grapes','mango','pomegranate','papaya','muskmelon','watermelon','apple','orange']
industrial = ['cotton','jute','coconut','coffee']
```

Final Approach: Random Forest Model

Why Model: Handles complex patterns easily

Features (11): N, K, P, soil_type, irrigation_frequency, water_source_type, ph, N_to_P, P_to_K, N_to_K, and

Crop_type_encoded (Cereal = 0, Fruit = 1, Industrial = 2, Legume = 3)

Target : Label (22 Crops)

Train/Test Split: 70/30

Max Depth: 12

Number of Trees: 300

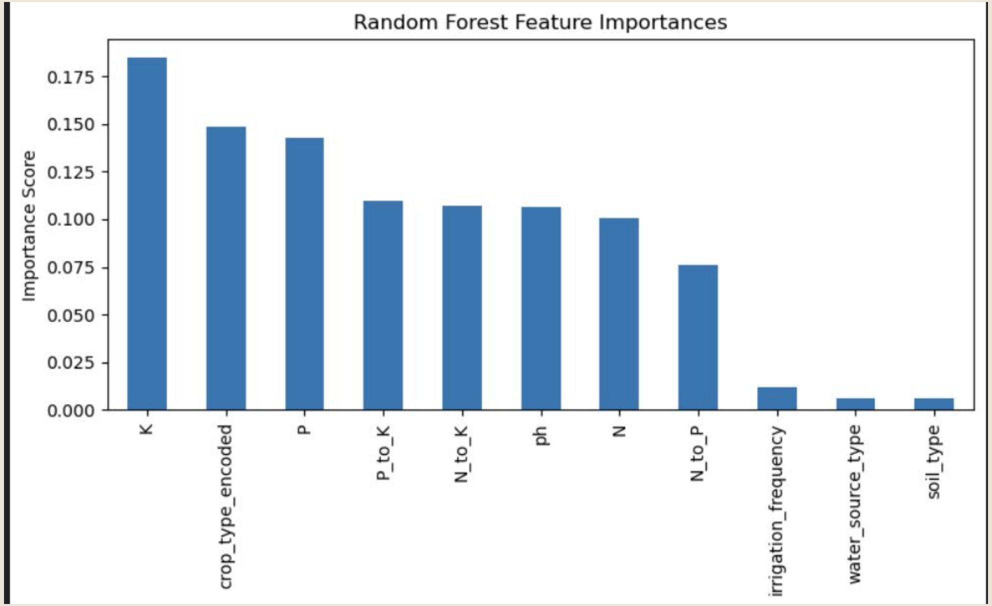
Random_State: 42

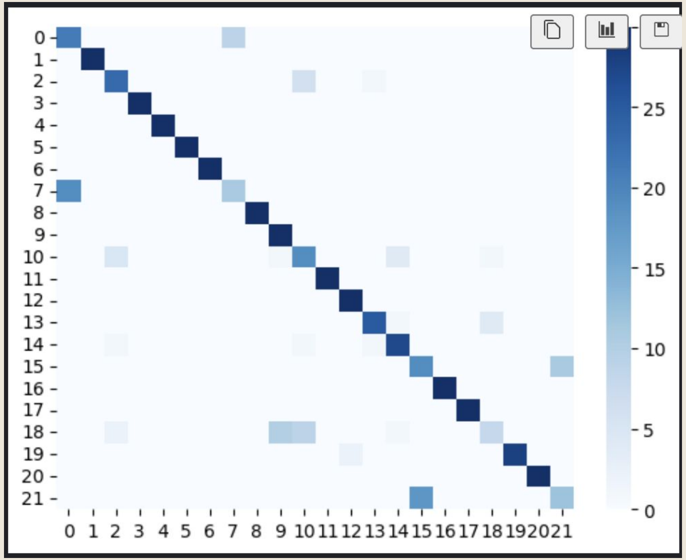
Accuracy: 0.8378787878787879

Classification Report:

	precision	recall	f1-score	support
apple	0.53	0.70	0.60	30
banana	1.00	1.00	1.00	30
blackgram	0.74	0.77	0.75	30
chickpea	1.00	1.00	1.00	30
coconut	1.00	1.00	1.00	30
coffee	1.00	1.00	1.00	30
cotton	1.00	1.00	1.00	30
grapes	0.55	0.37	0.44	30
jute	1.00	1.00	1.00	30
kidneybeans	0.73	1.00	0.85	30
lentil	0.54	0.63	0.58	30
maize	1.00	1.00	1.00	30
mango	0.94	1.00	0.97	30
mothbeans	0.93	0.83	0.88	30
mungbean	0.82	0.90	0.86	30
muskmelon	0.51	0.63	0.57	30
orange	1.00	1.00	1.00	30
papaya	1.00	1.00	1.00	30
pigeonpeas	0.62	0.27	0.37	30
pomegranate	1.00	0.93	0.97	30

Feature Importances and Findings





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Decision Tree

Decision Tree Regressor Results

- Target Variable = **logCropYield**
- Top Features:
    - growth_stage
    - WAI
    - rainfall,
    - crop_density
    - PP
    - THI
    - Fertilizer_usage

Results

- Model Performance
    - Best Depth: **10**
    - Best Min Samples Leaf: **6**
    - MSE: **0.22841910008232078**
    - R^2 Score: **0.7541096199784009**
    - Train/Test Split: **70/30**

```
Correlation with Log Crop Yield:
logCropYield                     1.000000
cropYield                        0.728928
estimatedAverageYieldPerPlant    0.623190
growth_stage                     0.469991
WAI                              0.423151
rainfall                         0.419329
crop_density                     0.409648
PP                               0.336589
sunlight_exposure                0.288333
temperature                      0.190734
temperatureInFahrenheit          0.190734
THI                              0.190203
co2_concentration                0.054454
soil_type                        0.049778
humidity                         0.043249
soil_moisture                    0.037997
irrigation_frequency             0.035399
N                                0.031006
water_source_type                0.028439
organic_matter                   0.027855
fertilizer_usage                 0.009403
wind_speed                      -0.000701
water_usage_efficiency          -0.005887
urban_area_proximity            -0.009668
SFI                             -0.035655
NBR                             -0.061827
ph                              -0.062936
P                               -0.064097
K                               -0.078414
frost_risk                      -0.364959
pest_pressure                   -0.371694
Name: logCropYield, dtype: float64
```

Can Crops Maintain Performance with Less Fertilizer

- Crop Yield = amount of crop harvester per unit area in kg/ha

Compared avg predicted yield with

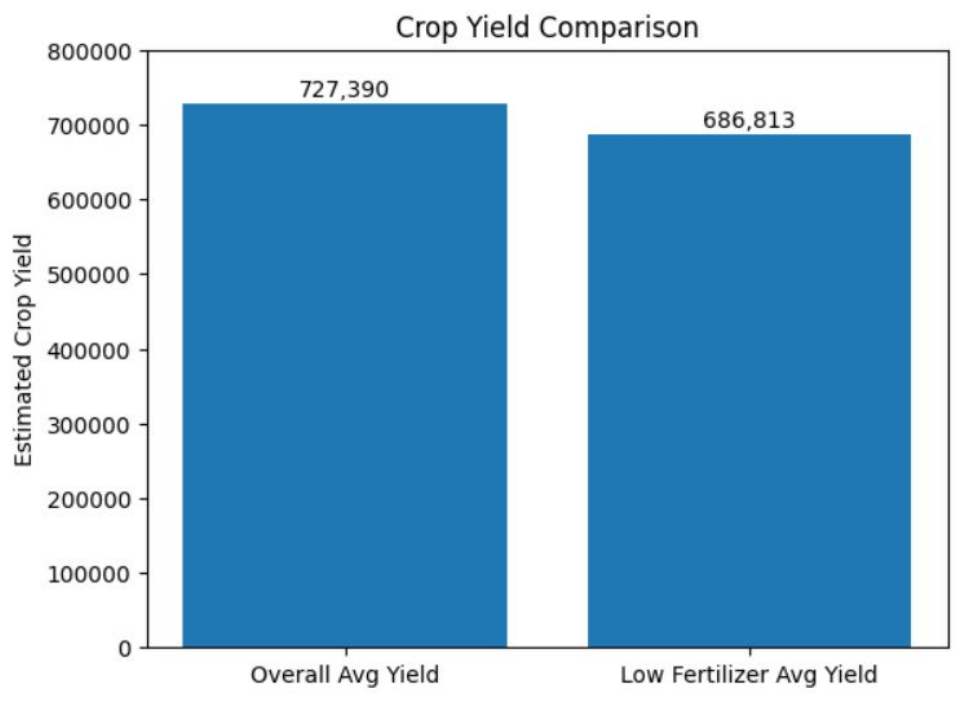
- Low fertilizer: **13.4409 → 687k**

Overall average Crop Yield:**13.4972 → 727k**

The difference 727,389 - 687,615 = **40,577**

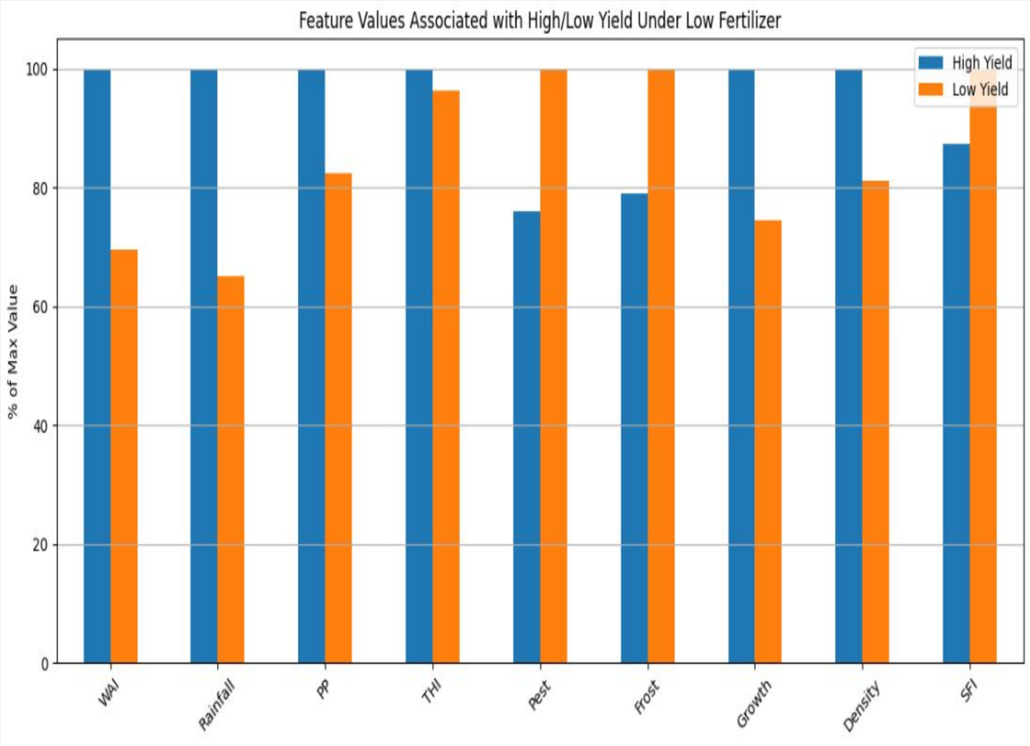
Conclusion:

- On average low fertilizer crops yield about 40,000 units less than the overall average which is roughly **5.6%**
- (40,577 / 727,390) * 100 ≈ **5.6%**
- **Crops with lower fertilizer can still perform well and competitively**



What Variables Allow Crops to Still Perform Well When Fertilizer is Low?

Feature	High Yield Avg Low Fertilizer Values	Low Yield Avg Low Fertilizer Values	Interpretation
WAI (Water Availability Index)	148.33	103.31	More water = higher yield
Rainfall	128.05	83.81	More rain = better performance under low fertilizer
Crop Density	13.49	10.95	With low fertilizer denser planting may improve yield
PP	95,601	78,798	With more sunlight and co2 = higher yield



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Ordinary Least Squares Regression

Which soil and environmental factors maximize water usage efficiency?

Predictor variables

- Soil moisture
- Soil type
- N, P, K
- Temperature
- Humidity
- ph
- Rainfall
- Organic matter
- Irrigation frequency
- 11 variables in original model

Target variable: water usage efficiency

- Full model performed really poorly
- R^2= 0.007

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     water_usage_efficiency   R-squared:                   0.007
Model:                                OLS   Adj. R-squared:              0.000
Method:                     Least Squares   F-statistic:                 1.015
Date:                    Wed, 19 Nov 2025   Prob (F-statistic):          0.430
Time:                            18:40:40   Log-Likelihood:            -2381.0
No. Observations:                    1540   AIC:                         4786.
Df Residuals:                        1528   BIC:                         4850.
Df Model:                              11
```

Backwards Selection

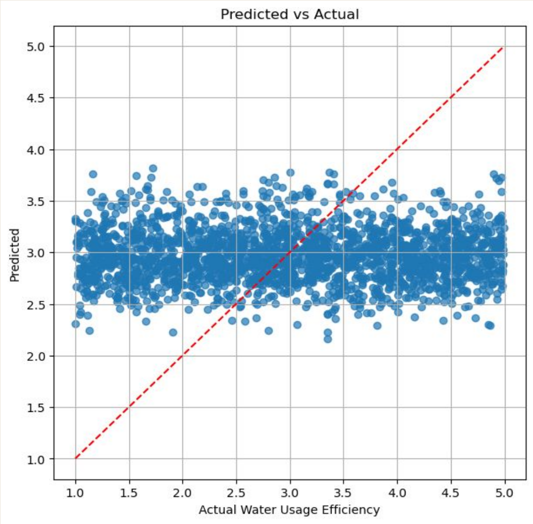
```
Final Features Selected:
Index(['soil_moisture', 'soil_type', 'P', 'temperature', 'ph', 'rainfall'], dtype='object')

Final Model Summary:
                                 OLS Regression Results
=======================================================================================
Dep. Variable:     water_usage_efficiency   R-squared (uncentered):                   0.867
Model:                                OLS   Adj. R-squared (uncentered):              0.867
Method:                     Least Squares   F-statistic:                              2387.
Date:                    Wed, 19 Nov 2025   Prob (F-statistic):                        0.00
Time:                            18:56:09   Log-Likelihood:                         -3460.1
No. Observations:                    2200   AIC:                                      6932.
Df Residuals:                        2194   BIC:                                      6966.
Df Model:                               6
Covariance Type:                nonrobust
```

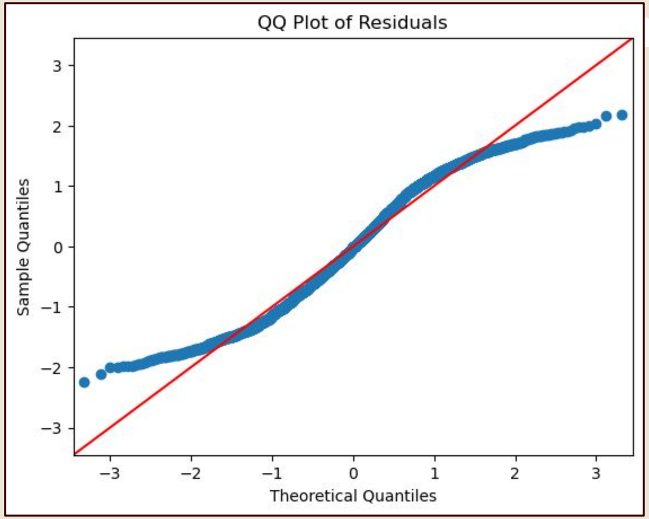
Final Variables Selected

- Soil moisture
- Soil type
- P
- Temperature
- ph
- rainfall

- Based on p-values
- New R^2: 0.867

Clear Normality and Linearity Violations



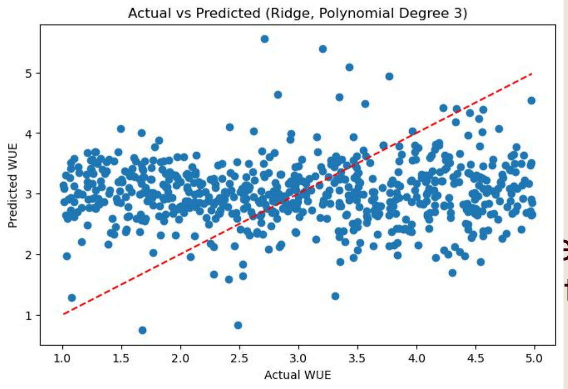


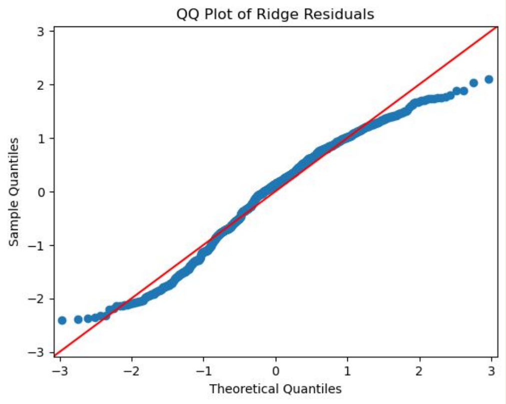
Polynomial regression

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     water_usage_efficiency   R-squared:                       0.031
Model:                                OLS   Adj. R-squared:                  0.001
Method:                     Least Squares   F-statistic:                     1.050
Date:                    Thu, 20 Nov 2025   Prob (F-statistic):              0.369
Time:                            11:38:24   Log-Likelihood:                -1237.7
No. Observations:                    2200   AIC:                             2607.
Df Residuals:                        2134   BIC:                             2983.
Df Model:                              65
Covariance Type:                nonrobust
```

- Log transformation of y variable
- Polynomial expansion
- Still a low R^2
- Plots didn't change

Ridge Regression






```
TRAIN RMSE: 0.39754666775717873
TEST RMSE: 0.46704486155350267
TRAIN R²: 0.15327945083403216
TEST R²: -0.17929939927941163
                                  Feature  Coefficient
9                          organic_matter     0.288860
6                                humidity    -0.255089
5                             temperature     0.231884
2                                       N    -0.217620
4                                       K    -0.147522
47                         temperature ph    -0.098775
56                                   ph^2    -0.088012
10                   irrigation_frequency     0.080869
8                                rainfall     0.057433
1                           soil_moisture     0.057185
52                            humidity ph     0.053833
58                      ph organic_matter    -0.052830
41                                   K ph     0.040074
64    organic_matter irrigation_frequency     0.029229
3                                       P    -0.026299
26                                   N ph     0.024030
65                 irrigation_frequency^2     0.023174
34                                   P ph     0.022382
57                            ph rainfall    -0.019052
20     soil_moisture irrigation_frequency    -0.017309
```

- Included polynomial expansion
- Slight improvement